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Semiconductor devices with flexible reinforcement structure

Abstract

Methods for manufacturing semiconductor devices having a flexible reinforcement structure, and associated systems and devices, are disclosed herein. In one embodiment, a method of manufacturing a semiconductor device includes electrically coupling at least one semiconductor die to a redistribution structure on a first carrier. The semiconductor die can include a first surface facing the redistribution structure and a second surface spaced apart from the redistribution structure. The method also includes reducing a thickness of the semiconductor die to no more than 10 μm . The method further includes coupling a flexible reinforcement structure to the second surface of the at least one semiconductor die.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/106,225, filed Feb. 6, 2023; which is a continuation of U.S. patent Ser. No. 16/896,043, filed Jun. 8, 2020, now U.S. Pat. No. 11,574,820; each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present technology generally relates to semiconductor devices, and more particularly relates to semiconductor devices having a flexible reinforcement structure coupled to the semiconductor die.

BACKGROUND

(2) Packaged semiconductor dies, including memory chips, microprocessor chips, and imager chips, typically include a semiconductor die mounted on a substrate and encased in a protective covering. The semiconductor die can include functional features, such as memory cells, processor circuits, and imager devices, as well as bond pads electrically connected to the functional features. The bond pads can be electrically connected to terminals outside the protective covering to allow the semiconductor die to be connected to higher level circuitry.

(3) During the manufacturing process, the semiconductor die can be thinned (e.g., by back grinding) to reduce the overall thickness of the semiconductor package. However, thinner semiconductor dies may be more prone to chipping, cracking, or other damage during subsequent processing steps. Semiconductor packages with thinner semiconductor dies may be also more prone to failure from thermomechanical stresses (e.g., chip-package interaction (CPI) stresses).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the present technology can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on illustrating clearly the principles of the present technology.

(2) FIG. 1A is a side cross-sectional view of a redistribution structure formed on a first carrier and configured in accordance with embodiments of the present technology.

(3) FIG. 1B is a side cross-sectional view of semiconductor dies coupled to the redistribution structure of FIG. 1A and configured in accordance with embodiments of the present technology.

(4) FIG. 1C is a side cross-sectional view of the semiconductor dies of FIG. 1B encapsulated in a mold material configured in accordance with embodiments of the present technology.

(5) FIG. 1D is a side cross-sectional view of the semiconductor dies of FIG. 1C after thinning in accordance with embodiments of the present technology.

(6) FIG. 1E is a side cross-sectional view of a reinforcement structure attached to the thinned semiconductor dies of FIG. 1D and configured in accordance with embodiments of the present technology.

- (7) FIG. 1F is a side cross-sectional view of a second carrier attached to the reinforcement structure of FIG. 1E and configured in accordance with embodiments of the present technology.
- (8) FIG. 1G is a side cross-sectional view of the semiconductor dies of FIG. 1F after separation from the first carrier in accordance with embodiments of the present technology.
- (9) FIG. 1H is a side cross-sectional view of the semiconductor dies of FIG. 1G coupled to an array of electrical connectors configured in accordance with embodiments of the present technology.
- (10) FIG. 1I is a side cross-sectional view of a plurality of semiconductor devices produced by singulating the semiconductor dies of FIG. 1H in accordance with embodiments of the present technology.
- (11) FIG. 2 is a side cross-sectional view of a semiconductor device configured in accordance with embodiments of the present technology.
- (12) FIG. 3 is a schematic view of a system that includes a semiconductor device or package configured in accordance with embodiments of the present technology.

DETAILED DESCRIPTION

(13) Specific details of several embodiments of semiconductor devices, and associated systems and methods, are described below. A person skilled in the relevant art will recognize that suitable stages of the methods described herein can be performed at the wafer level or at the die level. Therefore, depending upon the context in which it is used, the term “substrate” can refer to a wafer-level substrate or to a singulated, die-level substrate. Furthermore, unless the context indicates otherwise, structures disclosed herein can be formed using conventional semiconductor-manufacturing techniques. Materials can be deposited, for example, using chemical vapor deposition, physical vapor deposition, atomic layer deposition, plating, electroless plating, spin coating, and/or other suitable techniques. Similarly, materials can be removed, for example, using plasma etching, wet etching, chemical-mechanical planarization, or other suitable techniques.

(14) In several of the embodiments described below, a method of manufacturing a semiconductor device includes electrically coupling one or more semiconductor dies to a redistribution structure on a first carrier. Each semiconductor die can have a first surface (e.g., an active side or surface) connected to the redistribution structure and a second surface (e.g., a back side or surface) spaced apart from the redistribution structure. The method can also include reducing a thickness of each semiconductor die, such as by grinding or otherwise removing material from the second surface of the semiconductor die. After the thinning process, a flexible reinforcement structure can be coupled to the second surface of the semiconductor dies, and the flexible reinforcement structure can remain attached to the semiconductor dies during subsequent manufacturing steps and/or with the final semiconductor device. The present technology is expected to reduce chipping, cracking, or other damage or failures in the semiconductor dies contributing to yield loss during the manufacturing process. Additionally, the embodiments described herein can be used to reliably produce very thin semiconductor dies (e.g., no more than 10 μm thick) suitable for use in a single die package (SDP) or system in package (SiP) for flexible electronics applications or other applications where thin dies are desirable. In some embodiments, the techniques described herein can be used to produce substrate-less semiconductor packages (e.g., packages in which the semiconductor die is mounted directly to a printed circuit board or flexible circuit without any intermediate package substrate).

(15) Numerous specific details are disclosed herein to provide a thorough and enabling description of embodiments of the present technology. A person skilled in the art, however, will understand that the technology may have additional embodiments and that the technology may be practiced without several of the details of the embodiments described below with reference to FIGS. 1A-3. For example, some details of semiconductor devices and/or packages well known in the art have been omitted so as not to obscure the present technology. In general, it should be understood that various other devices and systems in addition to those specific embodiments disclosed herein may be within the scope of the present technology.

(16) As used herein, the terms “vertical,” “lateral,” “upper,” “lower,” “above,” and “below” can refer to relative directions or positions of features in the semiconductor devices in view of the orientation shown in the Figures. For example, “upper” or “uppermost” can refer to a feature positioned closer to the top of a page than another feature. These terms, however, should be construed broadly to include semiconductor devices having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

(17) FIGS. 1A-1I are side cross-sectional views illustrating various stages in a method of manufacturing a semiconductor device in accordance with embodiments of the present technology. A semiconductor device can be manufactured, for example, as a discrete device or as part of a larger wafer or panel. In wafer-level or panel-level manufacturing, multiple semiconductor dies are packaged on a wafer or panel before being singulated into a plurality of individual devices. Although FIGS. 1A-1I illustrate a manufacturing process involving five semiconductor dies, in practice the process can be scaled or otherwise adapted for any suitable number of semiconductor dies (e.g., a single semiconductor die, tens or hundreds of semiconductor dies, etc.).

(18) Referring to FIG. 1A, a redistribution structure **100** (shown schematically) is formed on a first carrier **102**. The redistribution structure **100** includes a first surface **103a** (e.g., a lower surface) connected to the first carrier **102** and a second surface **103b** (e.g., an upper surface) away from the first carrier **102**. The redistribution structure **100** can be fabricated directly on the first carrier **102** using any suitable additive manufacturing process, such as sputtering, physical vapor deposition (PVD), electroplating, lithography, etc.

(19) In some embodiments, the redistribution structure **100** is or includes a redistribution layer (RDL) configured for fan-out wafer-level packaging in accordance with techniques known to those of skill in the art. The redistribution structure **100** can include one or more layers of an insulating material **104** and one or more layers of conductive elements **106** (e.g., contacts, traces, pads, vias, etc.). The insulating material **104** can separate and electrically isolate the conductive elements **106** from each other. The insulating material **104** can be made of any suitable non-conductive dielectric material, such as parylene, polyimide, or low temperature chemical vapor deposition (CVD) materials (e.g., tetraethylorthosilicate (TEOS), silicon nitride, silicon oxide). The conductive elements **106** can be made of any suitable conductive material, such as one or more metals (e.g., copper, silver, titanium, tungsten, cobalt, nickel, platinum, etc.), metal-containing compositions (e.g., metal silicide, metal nitride, metal carbide, etc.), and/or conductively-doped semiconductor materials (e.g., conductively-doped silicon, conductively-doped germanium, etc.). In some embodiments, the redistribution structure **100** is configured to be flexible and can undergo a certain amount of deformation (e.g., elastic and/or plastic deformation) while remaining fully operational. For example, the insulating material **104** can be made of a flexible polyimide dielectric material and the conductive elements **106** can be made of thin copper or flexible silver paste.

(20) The first carrier **102** can be a wafer or other structure that temporarily provides mechanical support to the redistribution structure **100** and/or other semiconductor components for subsequent processing stages. The first carrier **102** can be formed from any suitable material, such as silicon, silicon-on-insulator, compound semiconductor materials (e.g., gallium nitride), glass, or quartz. In some embodiments, the first carrier **102** is temporarily coupled to the redistribution structure **100** via a release layer **108**. The release layer **108** can be configured to be selectively dissolved, debonded, or otherwise separated from the redistribution structure **100** upon application of a suitable stimulus (e.g., heat, light) or agent (e.g., a solvent, an acid, water) so that the redistribution structure **100** can be removed from the first carrier **102**, as described in greater detail below. The release layer **108** can be made of any suitable material, such as an adhesive, a polymer, an epoxy, a film, a tape, a paste, etc.

(21) Referring to FIG. 1B, a plurality of semiconductor dies **110** are mechanically and electrically coupled to the redistribution structure **100**. The semiconductor dies **110** can be spaced apart from

each other across the surface of the redistribution structure **100**, e.g., distributed in accordance with wafer-level or panel-level manufacturing processes. Each semiconductor die **110** includes a semiconductor substrate **112** (e.g., a silicon substrate, a gallium arsenide substrate, an organic laminate substrate, etc.) having a first side or surface **114a** (e.g., a lower surface) and a second side or surface **114b** (e.g., an upper surface). Each semiconductor die **110** can be relatively thick, e.g., the initial thickness $T_{sub.1}$ of the semiconductor die **110** can be greater than or equal to 750 μm .

(22) In some embodiments, the first surface **114a** of each semiconductor die **110** is connected to the second surface **103b** of the redistribution structure **100** while the second surface **114b** is away from the redistribution structure **100**. The first surface **114a** can be an active side, surface, or region that includes various types of semiconductor components such as memory circuits, (e.g., dynamic random-access memory (DRAM), static random-access memory (SRAM), flash memory, or other type of memory circuits), controller circuits (e.g., DRAM controller circuits), logic circuits, processing circuits, circuit elements (e.g., wires, traces, interconnects, transistors, etc.), imaging components, and/or other semiconductor features. In some embodiments, the first surface **114a** includes contacts (e.g., bond pads—not shown) for electrically coupling the semiconductor die **110** to the conductive elements **106** of the redistribution structure **100**. The first surface **114a** can be electrically coupled to the redistribution structure **100** using any suitable bonding technique known to those of skill in the art (e.g., thermo-compression bonding, flip-chip bonding, etc.). The second surface **114b** can be a back side or surface of the semiconductor die **110** that does not include electrically or optically active semiconductor components or features.

(23) Referring to FIG. 1C, a mold material **116** can be disposed over the semiconductor dies **110** and portions of the second surface **103b** of the redistribution structure **100**. The mold material **116** can cover the second surface **114b** and lateral surfaces **114c** of each semiconductor die **110**, as well as fill the spaces between individual semiconductor dies **110**. The mold material **116** can be a resin, epoxy resin, silicone-based material, polyimide, or any other material suitable for encapsulating the semiconductor dies **110** and/or at least a portion of the redistribution structure **100** to protect these components from contaminants and/or physical damage. Once deposited, the mold material **116** can optionally be cured by UV light, chemical hardeners, heat, or other suitable curing methods known in the art.

(24) FIG. 1D shows the assembly after the thickness of each semiconductor die **110** has been reduced by removing material from the semiconductor substrate **112**. In some embodiments, the thinning process can involve removing portions of the semiconductor substrate **112** that do not include any active semiconductor components (e.g., portions at or near the second surface **114b**) while leaving the portions of the semiconductor substrate **112** that include active components (e.g., portions at or near the first surface **114a**). The thinning process can also involve removing some of the mold material **116**. For example, portions of the mold material **116** over the second surface **114b** can be removed so that the second surface **114b** is exposed for thinning the semiconductor dies **110** and subsequent manufacturing processes. Thinning of the semiconductor dies **110** can be accomplished via techniques known to those of skill in the art such as grinding (e.g., back grinding of the second surface **114b**), dry etching, chemical etching, or chemical mechanical polishing (CMP).

(25) The final thickness $T_{sub.2}$ of the semiconductor die **110** can be significantly smaller than the initial thickness $T_{sub.1}$. For example, the final thickness $T_{sub.2}$ can be less than or equal to 50 μm , 40 μm , 30 μm , 20 μm , 15 μm , 10 μm , 5 μm , 4 μm , 3 μm , 2 μm , or 1 μm . In some embodiments, the final thickness $T_{sub.2}$ is less than or equal to 10%, 5%, 4%, 3%, 2%, 1.5%, 1%, or 0.5% of the initial thickness $T_{sub.1}$. Optionally, the final thickness $T_{sub.2}$ can be sufficiently thin so that the semiconductor dies **110** can bend or flex for use in flexible electronics applications, non-TSV die stacking technologies, and/or other applications where very thin semiconductor dies are desirable, as described further below.

(26) Referring to FIG. 1E, at least one reinforcement structure **118** is coupled to the second surface

114b of each semiconductor die **110**. The reinforcement structure **118** includes a first surface **120a** (e.g., a lower surface) that contacts the semiconductor dies **110** and a second surface **120b** (e.g., an upper surface) away from the semiconductor dies **110**. In some embodiments, the second surface **114b** of each semiconductor die **110** contacts and is covered by the reinforcement structure **118**, while the lateral surfaces **114c** are surrounded by the mold material **116**. The reinforcement structure **118** can be configured as a layer, a sheet, a film, or any other structure having a sufficiently large surface area to cover multiple semiconductor dies **110** in a wafer-level or panel-level manufacturing process. The reinforcement structure **118** can also cover the exposed areas of the mold material **116**. The reinforcement structure **118** can be configured to support and protect the semiconductor dies **110** after thinning, e.g., to reduce or prevent cracking or chipping, mitigate the likelihood of failure due to thermomechanical stress, etc. As previously described, thin semiconductor dies (e.g., having thicknesses less than or equal to 50 μm , 35 μm , 10 μm , or 5 μm) can be particularly prone to chipping or cracking at the edges of the die in subsequent manufacturing processes. Additionally, thermomechanical stresses (e.g., CPI stresses) in subsequent manufacturing processes and/or during operation can also have a larger impact on thin semiconductor dies. Accordingly, the reinforcement structure **118** can protect the semiconductor dies **110** to reduce yield loss due to chipping, cracking, and other failures.

(27) In some embodiments, the reinforcement structure **118** is flexible and can accommodate a certain degree of elastic and/or plastic deformation (e.g., bending) without fracturing, separating from the semiconductor die **110**, or other mechanical failure. As such, the reinforcement structure **118** can be a relatively thin structure with a thickness $T_{\text{sub.3}}$ less than or equal to 50 μm , 40 μm , 30 μm , 20 μm , 15 μm , 10 μm , 5 μm , 4 μm , 3 μm , 2 μm , or 1 μm . In some embodiments, the thickness $T_{\text{sub.3}}$ is less than or equal to the thickness $T_{\text{sub.2}}$ of the semiconductor dies **110** after thinning, e.g., thickness $T_{\text{sub.3}}$ is less than or equal to 100%, 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, or 10% of thickness $T_{\text{sub.2}}$. The reinforcement structure **118** can be configured to be more flexible than a layer of silicon having an equivalent thickness.

(28) The reinforcement structure **118** can be made of many different types of materials, such as a polymeric material (e.g., polyimide, polytetrafluoroethylene (PTFE)), a resin (e.g., epoxy resin), a laminate, a film (e.g., a die attach film), a metallic material (e.g., copper, aluminum), or a combination thereof. Optionally, the reinforcement structure **118** can be a composite material including at least one structural element (e.g., weave, fiber, particle, etc.) embedded in a matrix material (e.g., a polymer, a resin, etc.). The structural element(s) can provide mechanical strength and support, while the matrix material can surround and/or impregnate the structural element(s) to connect them to each other and/or provide flexibility. For example, the matrix material can be a polyimide or an epoxy resin, and the structural elements can be a carbon-based material (e.g., carbon weave or carbon fibers) or a glass-based material (e.g., glass weave or glass fibers). Optionally, the reinforcement structure **118** can be made of materials that are identical or generally similar to materials used in substrates for semiconductor devices (e.g., core materials used in printed circuit boards). Alternatively or in combination, the reinforcement structure **118** can be made of materials that are identical or generally similar to metals or metal alloys used in semiconductor packaging and leadframe manufacturing (e.g., a metal layer attached via a thin adhesive layer).

(29) The reinforcement structure **118** can be coupled to the semiconductor dies **110** in various ways. For example, the reinforcement structure **118** can be provided as a premade layer or sheet that is laminated or otherwise bonded onto the semiconductor dies **110** (e.g., via heating, curing, adhesives, etc.). As another example, the reinforcement structure **118** can be provided as a liquid or a semi-solid material that is coated onto the semiconductor dies **110** (e.g., by spin coating, spray coating, etc.). In a further example, the reinforcement structure **118** can be molded onto the semiconductor dies **110**. The reinforcement structure **118** can optionally be attached to the semiconductor dies **110** in an uncured state, then subsequently cured (e.g., by heat, light, chemical

agents, etc.). Suitable adhesion promoters can also be used to facilitate coupling of the reinforcement structure **118** to the semiconductor dies **110**. In some embodiments, the second surface **114b** of the semiconductor substrate **112** includes a native oxide layer, and the reinforcement structure **118** is attached to the oxide layer. In other embodiments the native oxide layer can be removed so that the reinforcement structure **118** is directly attached to the silicon of the semiconductor substrate **112**. Other techniques known to those of skill in the art for fabricating a thin layer of material on the semiconductor dies **110** can also be used.

(30) Although the illustrated embodiments show a single reinforcement structure **118**, in other embodiments a different number of reinforcement structures **118** can be used (e.g., two, three, four, five, or more). The reinforcement structures **118** can each be made of the same material, or some or all of the reinforcement structures **118** can be made of different materials to impart different properties to the composite reinforcement structure (e.g., strength, flexibility, heat transfer, etc.). Likewise, the reinforcement structures **118** can each have the same thickness, or some or all of the reinforcement structures **118** can have different thicknesses. In some embodiments, the combined thickness of all the reinforcement structures **118** is sufficiently thin to maintain flexibility, e.g., less than or equal to 50 μm , 40 μm , 30 μm , 20 μm , 15 μm , 10 μm , 5 μm , 4 μm , 3 μm , 2 μm , or 1 μm .

(31) Referring to FIG. 1F, a second carrier **122** can be coupled to the reinforcement structure **118**. The second carrier **122** can be coupled to the second surface **120b** of the reinforcement structure **118** away from the semiconductor dies **110**. The second carrier **122** can be identical or generally similar to the first carrier **102**, such as a wafer or other structure that temporarily provides mechanical support for subsequent processing stages. The second carrier **122** can be coupled directly to the flexible reinforcement structure **118**, or can be indirectly coupled via a release layer (e.g., identical or generally similar to the release layer **108**—not shown) in accordance with techniques known to those of skill in the art.

(32) Referring to FIG. 1G, the first carrier **102** can be separated from the redistribution structure **100**. In some embodiments, the first carrier **102** is separated by dissolving, debonding, or otherwise decoupling the release layer **108** from the redistribution structure **100**, e.g., using a suitable stimulus (e.g., heat, light) or agent (e.g., solvent, water). Following removal of the first carrier **102**, the semiconductor dies **110** and redistribution structure **100** remain coupled to the second carrier **122** via the reinforcement structure **118**, while the first surface **103a** of the redistribution structure **100** is exposed for subsequent manufacturing stages.

(33) FIG. 1H shows the assembly after an array of electrical connectors **124** (e.g., a ball grid array) has been mechanically and electrically coupled to the redistribution structure **100**. The electrical connectors **124** can include solder balls, conductive bumps, conductive pillars, conductive epoxies, and/or other suitable electrically conductive elements configured to electrically and mechanically couple the semiconductor dies **110** to a substrate or another device, in accordance with techniques known to those of skill in the art. In some embodiments, the electrical connectors **124** are electrically coupled to the conductive elements **106** of the redistribution structure **100** such that signals from the semiconductor dies **110** can be routed to the electrical connectors **124** via the redistribution structure **100**. For example, the first surface **103a** of the redistribution structure **100** can include contacts (e.g., bond pads—not shown) for receiving and coupling to the electrical connectors **124**, and the contacts can be electrically coupled to the corresponding contacts on the semiconductor dies **110** via the conductive elements **106**.

(34) Referring to FIG. 1I, the semiconductor dies **110** can be singulated or otherwise separated from each other to form a plurality of individual semiconductor devices **150**. The singulation process can involve separating the reinforcement structure **118** from the second carrier **122** (e.g., using techniques similar to those previously described with respect to the first carrier **102** and FIG. 1G). Subsequently, the semiconductor dies **110** can be mounted on a tape or film **126** (e.g., dicing tape) supported by a frame **128**. In the illustrated embodiment, the second surface **120b** of the reinforcement structure **118** is coupled to the tape **126** while the first surface **103a** of the

redistribution structure **100** is exposed. A dicing mechanism (e.g., a blade, saw, laser, etc.) can be used to cut through the portions of the redistribution structure **100**, mold material **116**, and reinforcement structure **118** between the individual semiconductor dies **110** to separate them from each other. The resulting singulated devices **150** can be subsequently separated from the tape **126** in accordance with techniques known to those of skill in the art.

(35) Optionally, during the singulation process, a film layer (e.g., an anisotropic conductive film (ACF)—not shown) can be coupled between the tape **126** and the reinforcement structure **118**, e.g., to facilitate adhesion. The film layer can extend continuously across all of the devices **150**, or can be precut into individual sections corresponding to the size of an individual device **150**.

(36) In some embodiments, after singulation, one or more additional reinforcement structures (not shown) can be coupled around the lateral surfaces **152** of each device **150** to provide further support and protection. The additional reinforcement structures can be identical or generally similar to the reinforcement structure **118**, and can be attached via laminating, coating, molding, or any other suitable technique.

(37) FIG. 2 is a side cross-sectional view of a semiconductor device **200** configured in accordance with embodiments of the present technology. The device **200** can be manufactured using any embodiments of the methods provided herein, such as the method described with respect to FIG. 1A-1I. The device **200** includes a semiconductor die **110** mounted on a substrate **202**. The components of the device **200** can be identical or generally similar to the corresponding components previously described with respect to FIGS. 1A-1I. For example, the semiconductor die **110** can be relatively thin (e.g., having a thickness less than or equal to 10 μm or 5 μm), and can have a first surface **114a** (e.g., an active side or surface) coupled to a redistribution structure **100** and a second surface **114b** (e.g., a back side or surface) coupled to a reinforcement structure **118** (e.g., a flexible reinforcement structure). The semiconductor die **110** can be electrically coupled to the substrate **202** via the redistribution structure **100** and an array of electrical connectors **124**, thereby permitting signal transmission between the semiconductor die **110** and the substrate **202**. Optionally, the semiconductor die **110** can be at least partially encapsulated by a mold material **116** between the reinforcement structure **118** and redistribution structure **100**.

(38) The substrate **202** can be any structure or component suitable for supporting the semiconductor die **110**. For example, the substrate **202** can be or include an interposer, such as a printed circuit board, a dielectric spacer, another semiconductor die (e.g., a logic die), or another suitable substrate. In some embodiments, the substrate **202** is a flexible circuit or other suitable flexible substrate. Accordingly, the device **200** can be used in flexible electronics applications. In such embodiments, the device **200** can be configured to bend or otherwise deform while remaining fully operational (e.g., without mechanical and/or electrical failures). Optionally, the device **200** can be a “substrate-less” package in which the substrate **202** is a circuit board-level substrate (e.g., a printed circuit board or flexible circuit), and the device **200** does not include any intermediate substrates (e.g., package-level substrates) between the semiconductor die **110** and the circuit board-level substrate.

(39) Although the embodiments herein describe semiconductor devices having a single semiconductor die, the present technology can also be applied to semiconductor devices having a plurality of semiconductor dies that are vertically arranged in a die stack. Some or all of the semiconductor dies in the die stack can be relatively thin (e.g., having a thickness less than or equal to 10 μm or 5 μm). For example, some or all of the semiconductor dies can be sufficiently thin to accommodate non-TSV die stacking technologies. In such embodiments, one or more reinforcement structures can be coupled to the uppermost semiconductor die in the stack to provide protection against chipping, cracking, and/or thermomechanical stresses, as previously described. The reinforcement structure(s) can be attached to the uppermost semiconductor die (e.g., the semiconductor die furthest away from the redistribution structure) after the die stack has been assembled on a redistribution structure, in accordance with techniques known to those of skill in

the art. Accordingly, the corresponding semiconductor device can include a first semiconductor die coupled to the reinforcement structure(s) and at least one second semiconductor die between the first semiconductor die and the redistribution structure.

(40) Any one of the semiconductor devices and/or packages having the features described above with reference to FIGS. 1A-2 can be incorporated into any of a myriad of larger and/or more complex systems, a representative example of which is system **300** shown schematically in FIG. 3. The system **300** can include a processor **302**, a memory **304** (e.g., SRAM, DRAM, flash, and/or other memory devices), input/output devices **306**, and/or other subsystems or components **308**. The semiconductor dies and/or packages described above with reference to FIGS. 1A-2 can be included in any of the elements shown in FIG. 3. The resulting system **300** can be configured to perform any of a wide variety of suitable computing, processing, storage, sensing, imaging, and/or other functions. Accordingly, representative examples of the system **300** include, without limitation, computers and/or other data processors, such as desktop computers, laptop computers, Internet appliances, hand-held devices (e.g., palm-top computers, wearable computers, cellular or mobile phones, personal digital assistants, music players, etc.), tablets, multi-processor systems, processor-based or programmable consumer electronics, network computers, and minicomputers. Additional representative examples of the system **300** include lights, cameras, vehicles, etc. With regard to these and other example, the system **300** can be housed in a single unit or distributed over multiple interconnected units, e.g., through a communication network. The components of the system **300** can accordingly include local and/or remote memory storage devices and any of a wide variety of suitable computer-readable media.

(41) From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but that various modifications may be made without deviating from the disclosure. Accordingly, the invention is not limited except as by the appended claims. Furthermore, certain aspects of the new technology described in the context of particular embodiments may also be combined or eliminated in other embodiments. Moreover, although advantages associated with certain embodiments of the new technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1. A method of manufacturing a semiconductor device, the method comprising: electrically coupling a semiconductor die to a redistribution structure on a first carrier, the semiconductor die including a first surface facing the redistribution structure and a second surface spaced apart from the redistribution structure; reducing a thickness of the semiconductor die to no more than 10 μm ; coupling a flexible reinforcement structure to the second surface of the semiconductor die; coupling the flexible reinforcement structure to a second carrier; and separating the redistribution structure from the first carrier.
2. The method of claim 1, further comprising: separating the flexible reinforcement structure from the second carrier; and electrically coupling the semiconductor die to a flexible circuit via an array of electrical connectors.
3. The method of claim 1 wherein the flexible reinforcement structure comprises a polymeric material, a resin, a laminate, or a film.
4. The method of claim 1 wherein the flexible reinforcement structure comprises a weave or a fiber, wherein the weave or the fiber are embedded in a matrix comprised of polymer or resin.
5. The method of claim 4, wherein the weave or the fiber are comprised of carbon or glass.
6. The method of claim 1, wherein the semiconductor die comprises a plurality of semiconductor

dies, the method further comprising: singulating the plurality of semiconductor dies into a plurality of individual semiconductor devices.

7. The method of claim 1, wherein the first carrier and the redistribution structure are coupled by a release layer between the first carrier and the redistribution structure, the method further comprising: separating the first carrier from the redistribution structure by dissolving the release layer or debonding the release layer from the redistribution structure.

8. The method of claim 1, further comprising: forming a mold material over the semiconductor die and the redistribution structure, wherein reducing the thickness of the semiconductor die includes removing a portion of the mold material to expose the second surface of the semiconductor die.

9. A method of manufacturing a semiconductor device, the method comprising: electrically coupling a first semiconductor die to a redistribution structure on a first carrier, the first semiconductor die including a first surface facing the redistribution structure and a second surface spaced apart from the redistribution structure; reducing a thickness of the first semiconductor die to no more than 10 μm ; coupling a flexible reinforcement structure to the second surface of the first semiconductor die; and coupling a second semiconductor die to the first surface of the first semiconductor die, wherein the second semiconductor die is positioned between the first semiconductor die and the redistribution structure.

10. The method of claim 9, further comprising: coupling the redistribution structure and the first semiconductor die to a flexible circuit.

11. The method of claim 9 wherein the flexible reinforcement structure comprises a weave or a fiber, wherein the weave or the fiber are embedded in a matrix comprised of polymer or resin.

12. The method of claim 9, wherein the first semiconductor die comprises a plurality of semiconductor dies, the method further comprising: singulating the plurality of semiconductor dies into a plurality of individual semiconductor devices.

13. The method of claim 9, wherein the first carrier and the redistribution structure are coupled by a release layer between the first carrier and the redistribution structure, the method further comprising: separating the first carrier from the redistribution structure by dissolving the release layer or debonding the release layer from the redistribution structure.

14. The method of claim 9, further comprising: forming a mold material over the first semiconductor die and the redistribution structure, wherein reducing the thickness of the first semiconductor die includes removing a portion of the mold material to expose the second surface of the first semiconductor die.

15. A method of manufacturing a semiconductor device, the method comprising: electrically coupling a semiconductor die to a redistribution structure on a first carrier, the semiconductor die including a first surface facing the redistribution structure and a second surface spaced apart from the redistribution structure; reducing a thickness of the semiconductor die to no more than 10 μm ; coupling a first flexible reinforcement structure to the second surface of the semiconductor die; and coupling a second flexible reinforcement structure to the first flexible reinforcement structure.

16. The method of claim 15, wherein the first flexible reinforcement structure comprises a weave or a fiber, wherein the weave or the fiber are embedded in a matrix comprised of polymer or resin.

17. The method of claim 15, wherein the semiconductor die comprises a plurality of semiconductor dies, the method further comprising: singulating the plurality of semiconductor dies into a plurality of individual semiconductor devices.

18. The method of claim 15, wherein the first carrier and the redistribution structure are coupled by a release layer between the first carrier and the redistribution structure, the method further comprising: separating the first carrier from the redistribution structure by dissolving the release layer or debonding the release layer from the redistribution structure.

19. The method of claim 15, further comprising: forming a mold material over the semiconductor die and the redistribution structure, wherein reducing the thickness of the semiconductor die includes removing a portion of the mold material to expose the second surface of the

semiconductor die.

20. The method of claim 15, wherein the first flexible reinforcement structure has a thickness of no more than 10 μm .
